

Analysis and Performance Verification of Latency-Critical Network Technologies for Autonomous Driving Environments

Network technologies enabling autonomous vehicles to react quickly even in critical moments

TSN

DetNet

5G URLLC

Problem Context: Why is Network Delay Dangerous?

 **Scenario:** Autonomous vehicles driving at constant intervals on a highway, the front vehicle suddenly brakes!

Following vehicles must receive brake signal instantly. A few ms delay = moving several cm to tens of cm = chain collision



Speed 100km/h

27.8m/sec movement
10ms delay = 28cm extra distance



Network Dependency

Autonomous driving controls
movement
with sensors + networks



Chain Collision Risk

One vehicle's slow reaction
spreads risk to the entire platoon

→ Network Stability & Speed = Autonomous Vehicle Safety Itself

Introduction to Three Core Technologies

01

TSN

Time-Sensitive Networking

 Dedicated lane for ambulances

Dedicated time reservation for critical data

Controls network schedule to prioritize emergency brake signals even among normal traffic

Core Goal

**Consistently fast transmission
(Deterministic Transmission)**

02

DetNet

Deterministic Networking

 Same package to multiple couriers

Guarantees uninterrupted transmission

Replicates critical messages via multiple paths → Guarantees arrival even if one fails

Core Goal

**Prevents message loss
even under packet loss**

03

5G URLLC

Ultra-Reliable Low-Latency Comm.

 Simultaneous reaction in wireless

Ultra-Reliable Wireless Comm.

Delivers wireless emergency signals at almost the same timing in platooning

Core Goal

**Stable simultaneous transmission
even in wireless environment**

Simulation Overview

✓ Normal Env Stable communication without packet loss

① Sim 1

Normal Env

Baseline vs TSN

🔍 Priority handling of Critical packets

? Are emergency signals prioritized?

③ Sim 3

Abnormal (Wireless)

Wireless Baseline vs URLLC

🔍 Wireless env simultaneous responsiveness

? Is simultaneous reaction possible wirelessly?

! Abnormal Env Packet loss & path failure

② Sim 2

Abnormal (Packet Loss)

TSN only vs TSN + DetNet

🔍 Preventing message loss during packet loss

? Is fast transmission enough?

④ Sim 4

Extreme Failure

Baseline vs TSN+DetNet+URLLC

🔍 Final safety verification of integrated tech

? Highest safety when applied integrated?

Sim 1: Baseline vs TSN (Normal Env)

Delay / Jitter Comparison

Metric	Baseline	TSN	Improvement
Avg Delay	0.628 ms	0.082 ms	↓ 87%
Delay SD	High	Low	↓ Stable
Avg Jitter	High	Low	↓ Reduced

**TSN Core: Not just fast, but
'consistently fast' predictable transmission**

10m, 15m: Both crash due to physical braking distance limits

20m~: Baseline crashes due to inconsistent reaction (jitter)

TSN secures safety with stable priority delivery

Collision Results by Vehicle Gap

Gap	Baseline	TSN
10m	💥 Crash	💥 Crash
15m	💥 Crash	💥 Crash
20m	💥 Crash	✅ Safe
25m	💥 Crash	✅ Safe
30m	💥 Crash	✅ Safe

✅ **Conclusion: TSN heavily reduces delay & jitter → Safe gap without crash from 20m+**

Sim 2: TSN vs TSN+DetNet (Packet Loss Env)

DetNet Concept: Replicates message over multiple paths → Guarantees arrival even if one fails

 "Entrusting the same package to multiple couriers at the same time"

Delay / Jitter Comparison

Metric	TSN only	TSN + DetNet
Avg Delay	Baseline	More Reduced
Delay SD	Baseline	↓ Reduced
Avg Jitter	Baseline	↓ ~35%

TSN: Fast and constant transmission

DetNet: Never-dropping transmission

→ Maximizes reliability when both combined

Collision Results by Vehicle Gap

Gap	TSN only	TSN+DetNet
15m	💣 Crash	💣 Crash
17m	💣 Crash	💣 Crash
19m	💣 Crash	💣 Crash
21m	💣 Crash	✅ Safe
23m	💣 Crash	✅ Safe
25m	💣 Crash	✅ Safe

✅ Conclusion: DetNet's path replication directly prevents crashes from 21m+ in packet loss env

Sim 3: Wireless Baseline vs URLLC (Wireless Packet Loss Env)

URLLC: Ultra-Reliable Low-Latency Communication — 5G-based ultra-reliable & ultra-low latency wireless comm.

Designed for all vehicles to receive emergency signals almost identically in a platooning scenario

Wireless Environment Challenges

! Signal interference & jamming

! Distance changes by vehicle movement

! Unstable radio environment

✗ Baseline: No compensation mechanism

URLLC is not just 'fast 5G'.
It is a core safety technology that stably synchronizes
vehicle reaction timings even in wireless env

Collision Results by Vehicle Gap

Gap	Baseline	URLLC
15m	💥 Crash	💥 Crash
17m	💥 Crash	💥 Crash
19m	💥 Crash	💥 Crash
21m	💥 Crash	💥 Crash
23m	💥 Crash	✅ Safe
25m	💥 Crash	✅ Safe

✅ Conclusion: Baseline crashes at 23m+, URLLC is safe — Direct confirmation of wireless platooning safety

Sim 4: Baseline vs TSN+DetNet+URLLC Integrated (Extreme Failure Env)

TSN: Priority fast delivery

DetNet: Packet loss recovery

URLLC: Wireless simultaneous reception

↔ Three technologies cooperate to maximize overall autonomous network stability

🚗 Collision Results in Extreme Failure Env

Vehicle Gap	Baseline	TSN+DetNet+URLLC	Note
15m	💣 Crash	💣 Crash	Physical limits
17m	💣 Crash	💣 Crash	Physical limits
19m	💣 Crash	💣 Crash	Physical limits
21m	💣 Crash	💣 Crash	Physical limits
23m	💣 Crash	✅ Safe	Network tech diffs
25m	💣 Crash	✅ Safe	Network tech diffs

✅ Conclusion: Integrated 3 tech achieves safety from 23m+ even in extreme failure env

Conclusion & Summary



TSN

Fast & constantly delivers Critical messages → Better predictable vehicle reactions



DetNet

Guarantees no message loss during packet loss → Secures reliability



URLLC

Simultaneous stable reaction in wireless env → Wireless platooning safety



Integrated

Combines 3 tech for high safety & reliability under extreme network failure

Latency-Critical Network Tech is not just a performance enhancement, but a core safety tech to prevent crashes

Thank You

Group 8 | Analysis and Performance Verification of Latency-Critical Network Technologies for Autonomous Driving Environments

TSN

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Autonomous Safety